Unit Test Report

Project: Task Manager Application

Version: 2.0 | Section: J | Batch: 12 | Date: 05-11-2026 | Build under test: `v1.0.1` (Sprint 3)

Authors

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Revision History

Version	Date	Author	Change Summary
1.0	04-11-2026	Student Team	Results for tag v1.0: 139 passed, 4 expected failures.
2.0	05-11-2026	Student Team	Results for v1.0.1 after the Sprint 3 defect fixes.

Approvals

Role	Name	Signature / Date
QA Lead	Tharunyah Sudhakar	
Course Coordinator		

1. Result

Tests run	Passed	Failed	Expected failures (open defects)	Skipped	Pass rate
155	155	0	0	0	100.0%

All 155 tests ran: 155 passed, 0 failed, and 0 were expected failures. The 121 sub-checks inside them also passed. At v1.0, 4 tests were expected failures because of open defects. Sprint 3 fixed those defects, so they now pass as regression tests (section 4).

2. Scope and method

- **Level:** module level, as the Test Plan (section 5) defines unit tests. Most tests call a module through its API with Flask's test client, so they also exercise that module's database access. A few call functions directly.
- **Isolation:** every test gets a new temporary SQLite database.
- **Tool:** pytest, with `unittest`-style test classes (deviation D5).
- **Environment:** Python 3.13.7, Flask 3, SQLite. GitHub Actions is set up to repeat the suite on Python 3.11, 3.12 and 3.13.
- **Evidence:** `docs/testing/evidence/junit.xml` (run 2026-11-05T09:03:43, 54.3 s).

```
pytest --cov=app --cov-branch --junitxml=docs/testing/evidence/junit.xml
```

3. Results by module

Module	Test files	Tests	Passed	Failed	Expected failures
Access Control	test_access_control, test_authorization	4	4	0	0
Administration	test_admin	8	8	0	0
Application skeleton	test_app	4	4	0	0
Authentication	test_auth, test_invalid_login, test_passwords, test_sessions	19	19	0	0
Scheduling	test_categories, test_due_dates, test_priority, test_reminders, test_status	22	22	0	0
Dashboard	test_dashboard	4	4	0	0
Security	test_data_protection, test_https, test_security_logging	17	17	0	0
Reliability	test_health	4	4	0	0
Input validation (Sprint 3)	test_input_robustness	12	12	0	0
Database	test_integrity	4	4	0	0
Notification	test_notifications	10	10	0	0
Part 3 additions	test_part3_additions	22	22	0	0
Performance	test_performance	3	3	0	0
Search	test_search	6	6	0	0
Task	test_task_lifecycle, test_tasks	11	11	0	0
User Interface	test_ux	5	5	0	0

4. Defects fixed in Sprint 3

Defect	Severity	Summary	Fix	Status
DEF-01	Low	Very large integer in a URL or query gave HTTP 500	One bounded <int> URL converter for the whole app, so such ids give the normal 404. An offset that is too large gets 400.	Fixed in v1.0.1
DEF-02	Low	Date at the edge of the supported range gave HTTP 500	clean_optional_datetime catches OverflowError and refuses dates before 1900 with the normal 400.	Fixed in v1.0.1
DEF-03	Low	minutesBefore of 1e999 (infinity) gave HTTP 500	OverflowError added to the except clause, giving the normal 400.	Fixed in v1.0.1
DEF-04	Low	Lone Unicode surrogate in text gave HTTP 500	get_json_body, which every route uses, refuses such text with 400. Emoji and other valid text are still accepted (SV-36l).	Fixed in v1.0.1
DEF-05	Low	Very deeply nested JSON gave HTTP 500	get_json_body catches RecursionError and returns 400.	Fixed in v1.0.1

Severities are proposed by the test run and need the QA Lead's confirmation. `tests/test_input_robustness.py` sweeps every API route, so a route added later is checked automatically. Run against the v1.0 code, these tests fail with the same errors; the output is in `docs/testing/evidence/v1.0/new_tests_against_v1.0_code.txt`.

5. All tests

Test file :: test	Test Case	Time (s)	Result
`test_access_control::test_TC_Sec_01_cannot_access_another_users_task`	TC-Sec-01	0.349	**PASS**

Test file :: test	Test Case	Time (s)	Result
test_admin::test_TC_Admin_01_only_admins_can_manage_users	TC-Admin-01	0.264	PASS
test_admin::test_TC_Sec_05b_admin_actions_logged	TC-Sec-05	0.207	PASS
test_admin::test_admin_cannot_lock_themselves_out	TC-Admin-01	0.196	PASS
test_admin::test_create_admin_cli	TC-Admin-01	0.196	PASS
test_admin::test_deactivation_logs_the_user_out_and_blocks_login	TC-Admin-01	0.246	PASS
test_admin::test_invalid_admin_updates_rejected	TC-Admin-01	0.200	PASS
test_admin::test_promote_and_delete_user	TC-Admin-01	0.222	PASS
test_admin::test_security_events_review	TC-Admin-01	0.178	PASS
test_app::test_database_schema_created	General	0.084	PASS
test_app::test_index_page_served	General	0.143	PASS
test_app::test_unexpected_errors_return_clean_json	General	0.090	PASS
test_app::test_unknown_api_route_returns_json_404	General	0.091	PASS
test_auth::test_TC_Auth_01_register_creates_unique_account	TC-Auth-01	0.101	PASS
test_auth::test_TC_Auth_01b_register_validates_input	TC-Auth-01	0.206	PASS
test_auth::test_register_rejects_non_json_body	TC-Val-01	0.090	PASS
test_auth::test_TC_Auth_02_valid_login_returns_session_token	TC-Auth-02	0.132	PASS
test_auth::test_login_email_is_case_insensitive	TC-Auth-02	0.114	PASS
test_auth::test_login_requires_both_fields	TC-Auth-03	0.091	PASS
test_authorization::test_TC_Sec_01b_all_api_routes_require_login	TC-Sec-01	0.121	PASS
test_authorization::test_public_endpoints_stay_public	TC-Sec-01	0.086	PASS
test_authorization::test_admin_required_checks_role	TC-Sec-01	0.170	PASS
test_categories::test_TC_Org_03_create_or_select_categories	TC-Org-03	0.183	PASS
test_categories::test_categories_are_private	TC-Org-03	0.184	PASS
test_categories::test_deleting_a_category_keeps_its_tasks	TC-Org-03	0.150	PASS
test_categories::test_edit_replaces_categories	TC-Org-03	0.167	PASS
test_categories::test_invalid_categories_rejected	TC-Org-03	0.133	PASS
test_dashboard::test_TC_Progress_01_counts_match_task_data	TC-Progress-01	0.454	PASS
test_dashboard::test_due_soon_lists_open_tasks_due_this_week	TC-Progress-01	0.208	PASS
test_dashboard::test_empty_dashboard	TC-Progress-01	0.125	PASS
test_dashboard::test_other_users_tasks_not_counted	TC-Progress-01	0.191	PASS
test_data_protection::test_TC_Sec_02b_security_headers_present	TC-Sec-02	0.106	PASS
test_data_protection::test_api_responses_are_never_cached	TC-Sec-02	0.118	PASS
test_data_protection::test_html_in_task_text_is_stored_as_plain_text	TC-Sec-01	0.149	PASS
test_data_protection::test_password_hash_never_leaves_the_server	TC-Sec-02	0.180	PASS
test_data_protection::test_search_input_cannot_inject_sql	TC-Sec-02	0.189	PASS
test_due_dates::test_TC_Org_04_due_date_stored_and_displayed	TC-Org-04	0.162	PASS
test_due_dates::test_due_date_can_be_changed_and_cleared	TC-Org-04	0.190	PASS

Test file :: test	Test Case	Time (s)	Result
test_due_dates::test_due_date_change_is_recorded_in_history	TC-Org-04	0.155	PASS
test_due_dates::test_invalid_due_date_rejected	TC-Org-04	0.160	PASS
test_health::test_TC_Reliability_01_health_endpoint	TC-Reliability-01	0.081	PASS
test_health::test_app_recovers_after_restart	TC-Reliability-01	0.164	PASS
test_health::test_health_reports_database_failure	TC-Reliability-01	0.087	PASS
test_health::test_health_reveals_nothing_private	TC-Reliability-01	0.131	PASS
test_https::test_TC_Sec_03_http_redirected_to_https_with_hsts	TC-Sec-03	0.087	PASS
test_https::test_login_over_plain_http_never_reaches_the_app	TC-Sec-03	0.084	PASS
test_https::test_proxy_header_trusted_only_when_configured	TC-Sec-03	0.081	PASS
test_https::test_forwarded_https_accepted	TC-Sec-03	0.080	PASS
test_https::test_production_forces_https_development_does_not	TC-Sec-03	0.083	PASS
test_input_robustness::test_TC_Val_01b_huge_id_in_any_url_is_not_found	TC-Val-01	0.182	PASS
test_input_robustness::test_TC_Val_01c_largest_valid_id_is_an_ordinary_not_found	TC-Val-01	0.159	PASS
test_input_robustness::test_TC_Val_01d_surrogate_in_any_json_body_is_refused	TC-Val-01	0.280	PASS
test_input_robustness::test_TC_Val_01e_deeply_nested_json_is_refused	TC-Val-01	0.156	PASS
test_input_robustness::test_TC_Val_01f_huge_paging_values_are_refused_or_clamped	TC-Val-01	0.173	PASS
test_input_robustness::test_TC_Val_01g_dates_outside_the_supported_range_are_refused	TC-Val-01	0.199	PASS
test_input_robustness::test_TC_Val_01h_dates_at_the_edges_of_the_range_are_accepted	TC-Val-01	0.166	PASS
test_input_robustness::test_TC_Val_01i_reminder_before_the_earliest_date_is_refused_cleanly	TC-Val-01	0.162	PASS
test_input_robustness::test_TC_Val_01j_infinite_or_huge_minutes_before_is_refused	TC-Val-01	0.165	PASS
test_input_robustness::test_TC_Val_01k_surrogate_in_login_and_registration_is_refused	TC-Val-01	0.145	PASS
test_input_robustness::test_TC_Val_01l_valid_unicode_is_still_accepted	TC-Val-01	0.168	PASS
test_input_robustness::test_TC_Val_01m_seeded_fuzz_of_every_route_gives_no_server_error	TC-Val-01	11.444	PASS
test_integrity::test_TC_Integrity_01_repeated_crud_preserves_state	TC-Integrity-01	1.229	PASS
test_integrity::test_crash_while_saving_a_task_leaves_nothing_behind	TC-Integrity-01	0.128	PASS
test_integrity::test_failed_transaction_rolls_back	TC-Integrity-01	0.135	PASS
test_integrity::test_invalid_edit_changes_nothing	TC-Integrity-01	0.126	PASS
test_invalid_login::test_TC_Auth_03_invalid_credentials_rejected	TC-Auth-03	0.132	PASS
test_invalid_login::test_deactivated_account_cannot_log_in	TC-Admin-01	0.136	PASS
test_invalid_login::test_unknown_email_still_checks_a_password_hash	TC-Auth-03	0.096	PASS

Test file :: test	Test Case	Time (s)	Result
test_notifications::test_TC_Notify_01_notification_generated_for_due_reminder	TC-Notify-01	0.159	PASS
test_notifications::test_TC_Org_05b_reminder_triggered_once_at_its_time	TC-Org-05	0.165	PASS
test_notifications::test_completed_tasks_do_not_notify	TC-Notify-01	0.162	PASS
test_notifications::test_lazy_trigger_when_listing_notifications	TC-Notify-01	0.157	PASS
test_notifications::test_notifications_are_private	TC-Notify-01	0.189	PASS
test_notifications::test_send_reminders_cli	TC-Notify-01	0.159	PASS
test_notifications::test_TC_Notify_02_preferences_apply_to_later_notifications	TC-Notify-02	0.212	PASS
test_notifications::test_defaults_before_anything_is_saved	TC-Notify-02	0.121	PASS
test_notifications::test_invalid_preferences_rejected	TC-Notify-02	0.144	PASS
test_notifications::test_task_preferences_apply_to_new_tasks_and_the_list	TC-Notify-02	0.173	PASS
test_part3_additions::test_TC_Admin_01b_create_admin_command_reports_invalid_input	TC-Admin-01	0.097	PASS
test_part3_additions::test_TC_Integrity_01b_init_db_command_is_safe_to_repeat	TC-Integrity-01	0.170	PASS
test_part3_additions::test_TC_Notify_01b_background_thread_delivers_due_reminders	TC-Notify-01	0.167	PASS
test_part3_additions::test_TC_Notify_01c_background_thread_survives_a_failed_pass	TC-Notify-01	0.090	PASS
test_part3_additions::test_TC_Notify_01d_background_thread_pass_with_nothing_due	TC-Notify-01	0.134	PASS
test_part3_additions::test_TC_Notify_01e_email_not_sent_when_the_service_is_disabled	TC-Notify-01	0.170	PASS
test_part3_additions::test_TC_Notify_01f_email_handed_to_the_service_when_enabled	TC-Notify-01	0.172	PASS
test_part3_additions::test_TC_Notify_01g_email_interface_returns_whether_it_was_sent	TC-Notify-01	0.087	PASS
test_part3_additions::test_TC_Notify_01i_mark_all_as_read	TC-Notify-01	0.211	PASS
test_part3_additions::test_TC_Sec_05c_admin_reads_all_events_and_bad_limit_is_refused	TC-Sec-05	0.137	PASS
test_part3_additions::test_TC_Auth_01c_name_longer_than_100_characters_refused	TC-Auth-01	0.080	PASS
test_part3_additions::test_TC_Org_03b_blank_category_names_are_dropped	TC-Org-03	0.152	PASS
test_part3_additions::test_TC_Org_04b_time_without_offset_is_taken_as_utc	TC-Org-04	0.087	PASS
test_part3_additions::test_TC_Task_01c_description_must_be_text_of_limited_length	TC-Task-01	0.144	PASS
test_part3_additions::test_TC_Task_02c_patch_without_real_change_writes_nothing	TC-Task-02	0.130	PASS
test_part3_additions::test_TC_UX_01b_unknown_page_keeps_the_normal_not_found_page	TC-UX-01	0.096	PASS
test_part3_additions::test_TC_Sec_04b_revoke_all_sessions_without_a_caller_transaction	TC-Sec-04	0.285	PASS
test_part3_additions::test_TC_Task_01d_plain_task_row_falls_back_to_the_stored_status	TC-Task-01	0.135	PASS

Test file :: test	Test Case	Time (s)	Result
test_part3_additions::test_DEF_01_huge_integer_in_url_or_query_is_not_a_server_error	TC-Val-01	0.109	PASS
test_part3_additions::test_DEF_02_date_outside_the_supported_range_is_not_a_server_error	TC-Val-01	0.140	PASS
test_part3_additions::test_DEF_03_infinite_minutes_before_is_not_a_server_error	TC-Val-01	0.223	PASS
test_part3_additions::test_DEF_04_lone_surrogate_in_text_is_not_a_server_error	TC-Val-01	0.214	PASS
test_passwords::test_TC_Sec_02_passwords_stored_only_as_hashes	TC-Sec-02	0.171	PASS
test_passwords::test_production_uses_scrypt	TC-Sec-02	0.125	PASS
test_passwords::test_same_password_gets_different_hashes	TC-Sec-02	0.246	PASS
test_performance::test_TC_Perf_01_dashboard_p95_under_3_seconds	TC-Perf-01	6.773	PASS
test_performance::test_task_list_with_300_tasks_is_fast	TC-Perf-01	6.028	PASS
test_performance::test_task_queries_use_indexes	TC-Perf-01	0.108	PASS
test_priority::test_TC_Org_01_priority_saved_and_displayed	TC-Org-01	0.245	PASS
test_priority::test_invalid_priority_rejected	TC-Org-01	0.204	PASS
test_priority::test_priority_change_is_recorded_in_history	TC-Org-01	0.266	PASS
test_reminders::test_TC_Org_05_reminder_configured_and_stored	TC-Org-05	0.311	PASS
test_reminders::test_delete_reminder	TC-Org-05	0.277	PASS
test_reminders::test_invalid_reminders_rejected	TC-Org-05	0.220	PASS
test_reminders::test_minutes_before_needs_a_due_date	TC-Org-05	0.198	PASS
test_reminders::test_minutes_before_uses_the_due_date	TC-Org-05	0.182	PASS
test_reminders::test_reminders_deleted_with_their_task	TC-Org-05	0.235	PASS
test_search::test_TC_Search_01_keyword_matches_title_or_description	TC-Search-01	0.433	PASS
test_search::test_TC_Search_02_filters	TC-Search-02	0.327	PASS
test_search::test_TC_Search_03_sorting	TC-Search-03	0.380	PASS
test_search::test_invalid_filters_rejected	TC-Search-02	0.341	PASS
test_search::test_pagination	TC-Search-03	0.333	PASS
test_search::test_search_never_returns_other_users_tasks	TC-Sec-01	0.354	PASS
test_security_logging::test_TC_Sec_05_event_logged_without_secrets	TC-Sec-05	0.148	PASS
test_security_logging::test_access_denied_is_logged	TC-Sec-05	0.270	PASS
test_security_logging::test_account_lifecycle_is_logged	TC-Sec-05	0.188	PASS
test_security_logging::test_failed_logins_logged_without_passwords	TC-Sec-05	0.220	PASS
test_security_logging::test_scrubber_removes_sensitive_keys	TC-Sec-05	0.146	PASS
test_security_logging::test_TC_Auth_03b_repeated_failures_are_rate_limited	TC-Auth-03	0.339	PASS
test_security_logging::test_lockout_ends_when_the_window_passes	TC-Auth-03	0.323	PASS
test_sessions::test_TC_Auth_02b_token_gives_access_to_protected_routes	TC-Auth-02	0.209	PASS
test_sessions::test_TC_Auth_04_logout_terminates_session	TC-Auth-04	0.253	PASS
test_sessions::test_TC_Sec_04_expired_and_forged_tokens_rejected	TC-Sec-04	0.236	PASS
test_sessions::test_deactivated_account_token_rejected	TC-Sec-04	0.214	PASS
test_sessions::test_logout_only_ends_that_session	TC-Sec-04	0.237	PASS

Test file :: test	Test Case	Time (s)	Result
test_sessions::test_missing_or_wrong_scheme_rejected	TC-Sec-04	0.185	PASS
test_sessions::test_update_profile	TC-Auth-02	0.355	PASS
test_status::test_TC_Org_02_status_assigned_and_updated	TC-Org-02	0.254	PASS
test_status::test_overdue_cannot_be_set_by_hand	TC-Org-02	0.277	PASS
test_status::test_overdue_is_derived_from_the_due_date	TC-Org-02	0.323	PASS
test_status::test_status_change_is_logged_in_history	TC-Org-02	0.181	PASS
test_task_lifecycle::test_TC_Task_03_delete_task	TC-Task-03	0.160	PASS
test_task_lifecycle::test_TC_Task_04_complete_and_reopen	TC-Task-04	0.137	PASS
test_task_lifecycle::test_completing_twice_is_harmless	TC-Task-04	0.120	PASS
test_task_lifecycle::test_delete_missing_task_returns_404	TC-Task-03	0.092	PASS
test_tasks::test_TC_Task_01_create_task_appears_in_list	TC-Task-01	0.105	PASS
test_tasks::test_TC_Task_01b_create_requires_title	TC-Task-01	0.135	PASS
test_tasks::test_TC_Task_02_edit_task	TC-Task-02	0.134	PASS
test_tasks::test_TC_Task_02b_invalid_field_rejects_whole_update	TC-Task-02	0.131	PASS
test_tasks::test_missing_task_returns_404	TC-Task-02	0.086	PASS
test_tasks::test_task_routes_require_login	TC-Sec-01	0.098	PASS
test_tasks::test_unknown_fields_rejected	TC-Task-02	0.113	PASS
test_ux::test_TC_UX_01_responsive_and_every_control_is_labelled	TC-UX-01	0.089	PASS
test_ux::test_dialogs_and_tables_are_named	TC-UX-01	0.076	PASS
test_ux::test_static_assets_served	TC-UX-01	0.070	PASS
test_ux::test_status_messages_are_announced_and_skip_link_exists	TC-UX-01	0.064	PASS
test_ux::test_text_colours_meet_wcag_aa_contrast_in_both_themes	TC-UX-01	0.063	PASS